Model Fit

Money Ballers

2025-11-23

```r
pacman::p_load(arrow, lme4, lmerTest, performance, broom, broom.mixed, patchwork,
               viridis, knitr, jtools, ggeffects, ggdist, gt, tidyverse)

options(scipen = 999)

# Set theme for plots
theme_set(theme_minimal())
```

```r
# Setting repository path
#repo_path <- "/Users/namomac/Desktop/Moneyball_FC/inputs/"
repo_path <- "D:/repos/Moneyball_FC/inputs/"

# Reading data
dat <- read_parquet(str_c(repo_path, "player_df.parquet"))
#dat <- read_parquet("player_df.parquet")
```

Time Centering

- **Year**: Centering at 2013=0 makes the intercept interpretable as the expected value at the study's start and avoids numerical issues that can arise with large year values (2013-2024). Also, including year^2 will allow us to study the (de)acceleration of the market over the decades.
  - Q: How does market value change annually compared to the 2013 baseline?

Grand Mean Centering

- **Age**: Market value is determined by biological career stage (e.g., "is Messi 29 and in his prime?" vs "is Messi 19 and a prospect?"). It does *not* matter if you are the oldest guy on a young team (relative age); the market still sees you as 29. We include the linear term (`age_c`) to catch the general trend and the squared term (`age_sq`) to capture the peak and decline curve.

  - Q: How does market value change as a player moves through their biological career arc (from prospect to prime to veteran)?

- **Foreign Player Proportion**: Since no club has 0% foreign players (lowest is 28%), we shift the zero point to the population average (Grand Mean) of internationalization. The intercept therefore represents a club with 'typical' diversity for the league.

  - Q: Does having a *higher proportion of foreign players than the average club* increase market value?

- **Height**: Height is a static physical trait. The market values a 195cm defender because he is objectively tall, not because he is taller than his teammates. Grand mean centering allows us to interpret the coefficient as "the premium for being 1cm taller than the average soccer player. Keeps the intercept interpretable as the value of an"average height" player.

  - Q: Does being taller than the average player in the league increase market value?

Year centering

- **UEFA** Coefficient = UEFA coefficients suffer from "grade inflation" or calculation changes over decades. A score of 15 in 2013 might mean something different than 15 in 2024. Centering by year isolates relative prestige, how dominant the club was that season compared to peers. It also removes the confounding effect of time-trends in the coefficient system itself.

  - Q: Does playing for a club that is currently outperforming the European average boost a player's value?

Year-Club mean centering (relative to teammates)

- **minutes_played_c**: Market value is driven by a player's status within their team. This metric isolates whether a player is a "Starter" (positive value) or a "Bench Player" (negative value) relative to their specific squad in that specific season. It removes the noise of "teams that rotate more" vs "teams that don't," focusing purely on the player's hierarchy in the squad.

  - Q: Does being a key player (playing more than the average teammate) increase market value?

Position centering:

- **Goals, Assists:** A goal scored by a striker is expected; a goal scored by a defender is rare and highly valuable. If we leave `goals` and `assists` as raw counts, the model treats 1 goal the same for everyone, potentially undervaluing defensive contributions. We will group by position as performance expectations vary entirely by role. When we center within the position group, we transform the variable from a raw count into a "performance above expectation" metric. This interprets as the effect of scoring above position expectations.
  - Q: Does scoring more goals than the average player in their position increase market value?" (This distinguishes a goal-scoring defender from an underperforming striker).

Non-centered fixed effects:

Performance Metrics

- **Red and Yellow card:** These variables have a meaningful 0 interpretation, and since our outcome is log transformed, these uncentered variables allow us to make percentage interpretations (i.e., one additional red card is associated with beta% change in market value).
  - Q: What is the direct market penalty for receiving one additional red/yellow card?
- **Position**: Reference group will be attackers.
  - Q: How much less is a defender valued compared to an attacker, holding performance constant?
- **Region**: Reference group European & Central Asia.
  - Q: What is the market premium for players in the 'Europe' region compared to South America?
- **Citizen**: Reference group is non-citizen of the country where the club is registered. The coefficient represents the simple difference in means between the two groups. A coefficient of +0.10 means "Citizens are valued 10% higher than Non-Citizens, all else held equal."
  - Q: Is there a homegrown premium' (or discount) for players who hold citizenship in the club's country compared to foreign players?

Random Intercept and slope

- Players: Yearly market values are nested within players so this makes them a clustering variable.
  - Q: After controlling for goals and age, how much variance in value is simply due to who the player is (marketing value/reputation)?
- Club: Players are nested within groups, but they can also be members of multiple clubs throughout their career making this a secondary clustering variable.
  - Q: How much of the variation in player value is due to the club they play for ('Real Madrid Effect')?
- Year: Without this random slope, the model assumes a one-size-fits-all growth rate (parallel lines). By including it, we allow for the possibility that some clubs are incubators (where values skyrocket year-over-year) while others are stagnant (where values grow slowly or decline), even after controlling for player performance. The theoretical explanation is that elite clubs (e.g., Real Madrid, Manchester City) benefit from massive commercial exposure and "brand halos" that cause their players' values to inflate faster than the market average. Conversely, smaller clubs may see their players' values strictly tied to performance without the "inflationary" bonus of the club's brand over time. Without this random slope, we may underestimate the standard errors for time varying fixed effects.
  - Q: Does the rate of growth in player market value differ significantly between clubs?

Why cross-classified model?

Players can change teams over their span of their careers meaning that their player context can change. They may be a star player in one team and a bench player on a different team that they switch to. In a cross-classified model, the player is the constant unit (i), but the club (j) changes over time (t). If player (I) plays for club (j1) in year (t0) but switches to club (j2) in year (t1) the fixed effect in the model, such as minutes played, updates the players status, as opposed to having static or average predictors. Club attributes, such as foreign players and UEFA scores, ensures that when player move clubs, the model swaps the values of X__j1, for X__j2. Player's characteristics, for instance age and height, move with the player across time.

Assumptions about the data:

We log transform market value to 2024 USD because financial data follows a pareto distribution rather than a normal distribution. If we keep the outcome on the raw scale, our residuals will be extremely large for high value players, and small for low value players violating the assumption of homoscedasticity. Log transforming the outcome compresses the extreme outliers by pulling the long right tail to try and make the distribution normal. We handle exchange rate fluctuation by using historical exchange rates to convert our values to USD.We also adjust market value to 2024 USD because 10 million in 2013 is not the same as in 2024 due to inflation, and by keeping the raw values we would lose real economic value. Therefore, this adjustment allows us to compare real economic power of a player in 2013 versus 2014 without the noise of currency fluctuation.

We assume that variables such as height and positions are time invariant for the duration of this study. At the club level, we assume mutually exclusiveness in that players can only belong to one club in a given year. Although a player could have switched clubs mid-season, we keep this constant for model simplicity. We assume that the relationship between age and log-market value is non-linear and parabolic so we introduce a quadratic term on age. Because we excluded observations with missing data (listwise deletion), we assume the data are Missing Completely at Random (MCAR) or that the proportion of missing cases (~2%) is negligible and does not introduce systematic bias.

Table 1: Data Structure for fixed effects, except year as it is a random slope

Variable Type	**Variable**	Centering Strategy/ Form Factor	Justification
Time	Season Year	Min-Centering (+ Quadratic)	Establishes a meaningful starting intercept (2013 =0)
Contextual, Club	UEFA Coefficient	Year-Mean Centering	Isolates relative prestige; removes historical inflation
	Foreign Players %	Grand-Mean Centering	Moves baseline to “Average Club” (since 0% is unlikely)
Player Characteristics	Age	Grand-Mean (+ Quadratic)	Captures absolute career stage (Prospect → Prime → Veteran)
	Height	Grand-Mean	Captures premium for absolute physical size
	Position	Reference group, attacker	Attackers tend to be valued more on average
Player Demographic	Citizen of Club Country	None (Raw binary 0/1)	Keeps “Non-Citizen” as clear reference group. Separates individual status from club composition (`foreign_players`)

Variable Type	Variable	Centering Strategy/ Form Factor	Justification
	Region of origin	Reference group, Europe	Control variable for where player is from
Player Performance	Goals	Position-Centering	Preserves natural 0 and marginal value of 1 goal interpretation
	Assists	Position-Centering	Preserves natural 0 and marginal value of 1 assist interpretation
	Red Card	raw	Meaningful 0 interpretation
	Yellow Card	raw	Meaningful 0 interpretation
	Average minutes played in season	Year-Mean Centering, z-score	Isolates whether player is a starter or bench player

```r
player_df <- dat |>
  mutate(
    player_id = factor(player_id),
    club_id   = factor(club_id),
    # reference groups
    position = fct_relevel(position, "Attack"),
    Region = fct_relevel(Region,  "Europe & Central Asia"),
    # outcome on log scale
    usd_revenue_2024_log = log(usd_revenue_2024),
    # season year
    # convert season_year to 0, 1, 2, etc.
    season_year_c = season_year - min(season_year),
    # grand mean centering
    foreign_players = foreign_players - mean(foreign_players, na.rm=TRUE),
    age_c = Age - mean(Age, na.rm=TRUE),
    height_c = height_in_cm - mean(height_in_cm, na.rm=TRUE)
  ) |>
  # year mean centering
```

Figure 1: Cross-Classified Data Structure: Players move between clubs over time.

```
  group_by(season_year_c) |>
  mutate(uefa_coefficient = uefa_coefficient - mean(uefa_coefficient, na.rm=TRUE),
         minutes_played = (minutes_played - mean(minutes_played, na.rm=TRUE)) /
           sd(minutes_played, na.rm = TRUE)) |>
  # position centering
  group_by(position) |>
  mutate(goals = goals - mean(goals, na.rm=TRUE),
         assists = assists - mean(assists, na.rm=TRUE)) |>
  ungroup()
```

- After checking missing values, we saw no more than 108 missing cases for Region, 103 for citizen of club country, and we decide to drop these cases as there are few and we are left with 7,023 cases.

```
# Which variable has missing value
player_df |>
  reframe(across(everything(), ~sum(is.na(.)))) |>
  select_if(~ . > 0) |>
  pivot_longer(everything(), names_to = "Variable", values_to = "Missing Cases") |>
  slice(1:5) |>
  arrange(desc(`Missing Cases`)) |>
  kable()
```

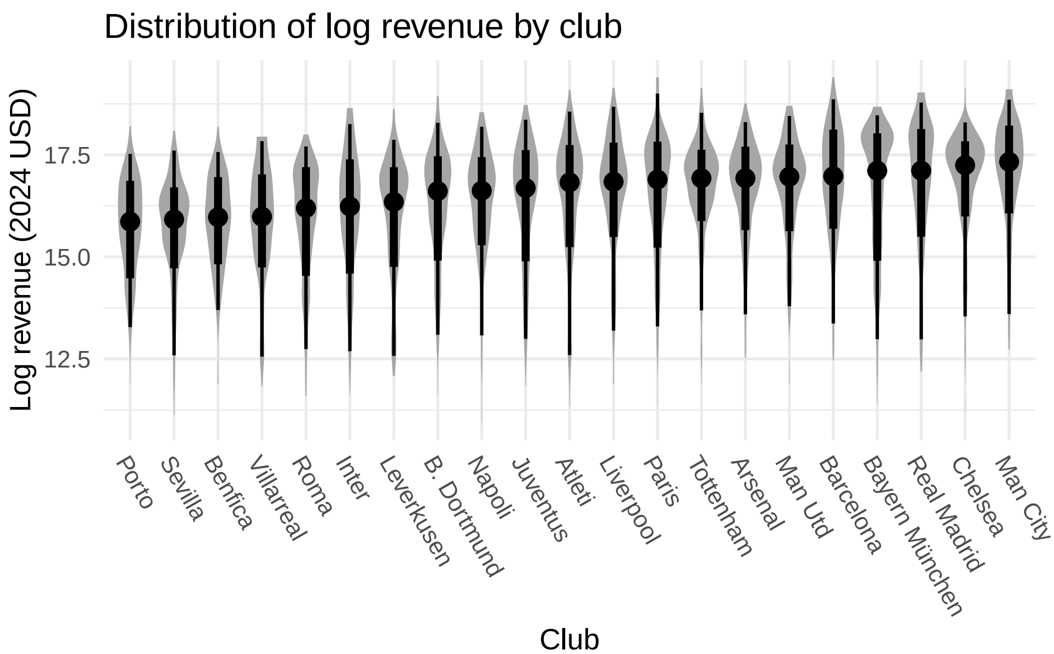
Variable	Missing Cases
country_of_birth	108
country_of_citizenship	103
height_in_cm	36
date_of_birth	2
Age	2

```
clean_player_df <- player_df |> drop_na()
```

- After examining the distribution of log market values across clubs, it is evident that the boxplots differ in both central tendency and variability. This indicates that club is a meaningful factor in explaining variation in market values.

```
clean_player_df |>
  mutate(Club = fct_reorder(factor(Club), usd_revenue_2024_log, median)) |>
  ggplot(aes(x=Club, y=usd_revenue_2024_log, group=Club)) +
  stat_slabinterval(side = "both")  +
```

```
  theme(axis.text.x=element_text(angle = -60, hjust = 0)) +
  labs(title = "Distribution of log revenue by club",
       y="Log revenue (2024 USD)")
```



Linearity

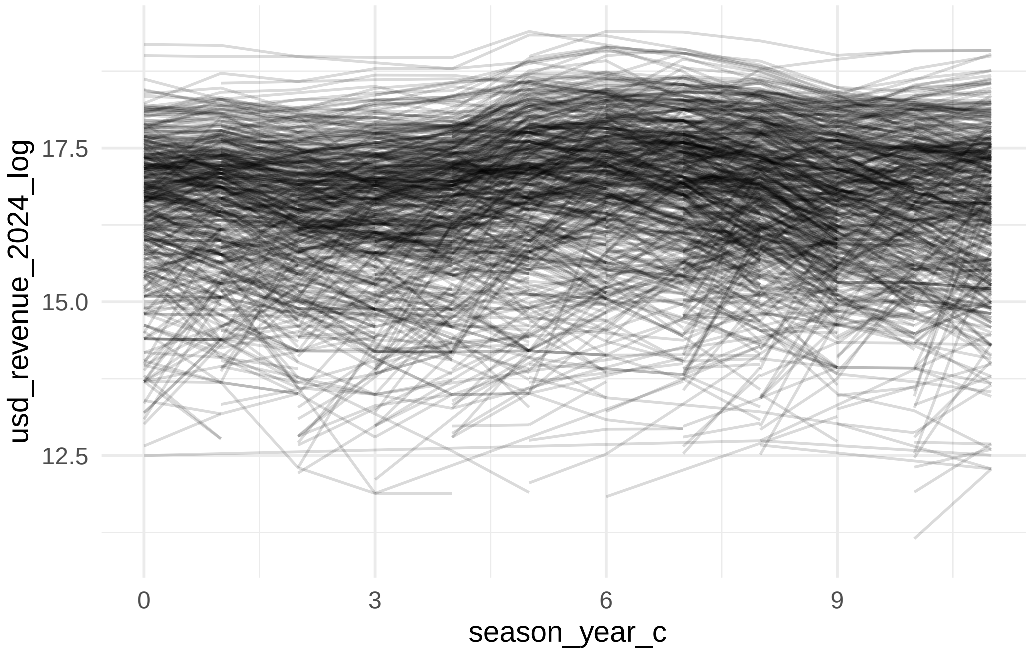
In the overall spaghetti plot group by play_id, the lines vary widely across players, showing considerable between-player heterogeneity in log market value. Within most players, the slopes appear mostly flat, but some of them change up and down dramatically.

In the overall spaghetti plot group by club_id, they are consistently show differences.This means between-club differences in market values are large, supporting the idea that club-level random intercepts are necessary in your multilevel model.

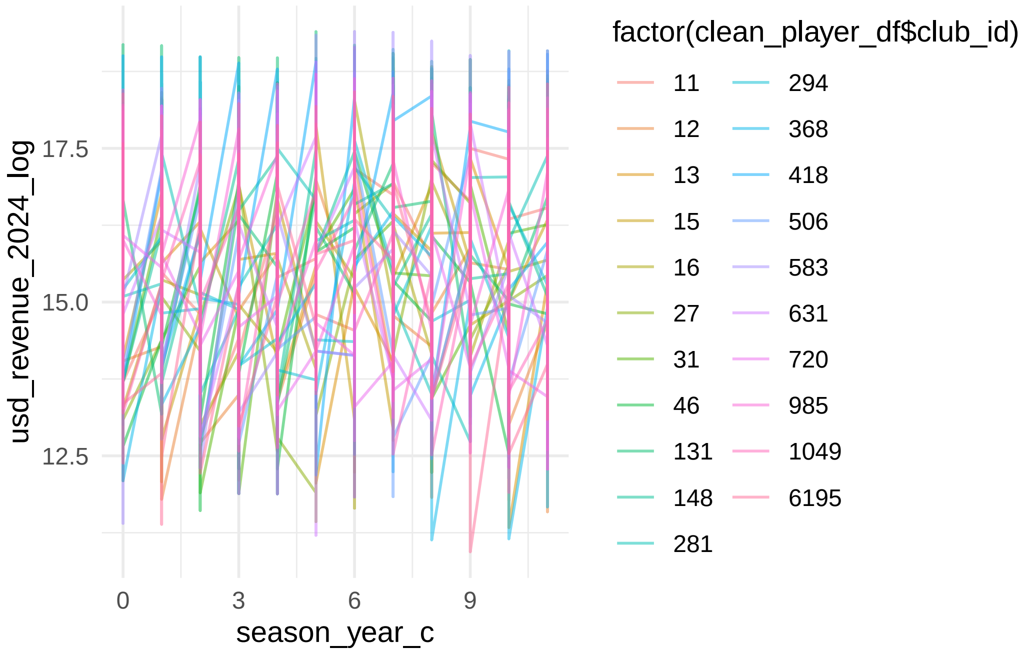
Besides,the clubs' average values change over time, indicating slopes might vary across clubs.

The loess smoother plot shows a non-linear averaged trend (curvature) in log-transformed market value (usd_revenue_2024_log) over centered season year (season_year_c). We can clearly see a peak at about season year 6 to 7, and the shape of the loess line is a wave pattern. Thus, we can try to add polynominal terms in our model to better explain the data.

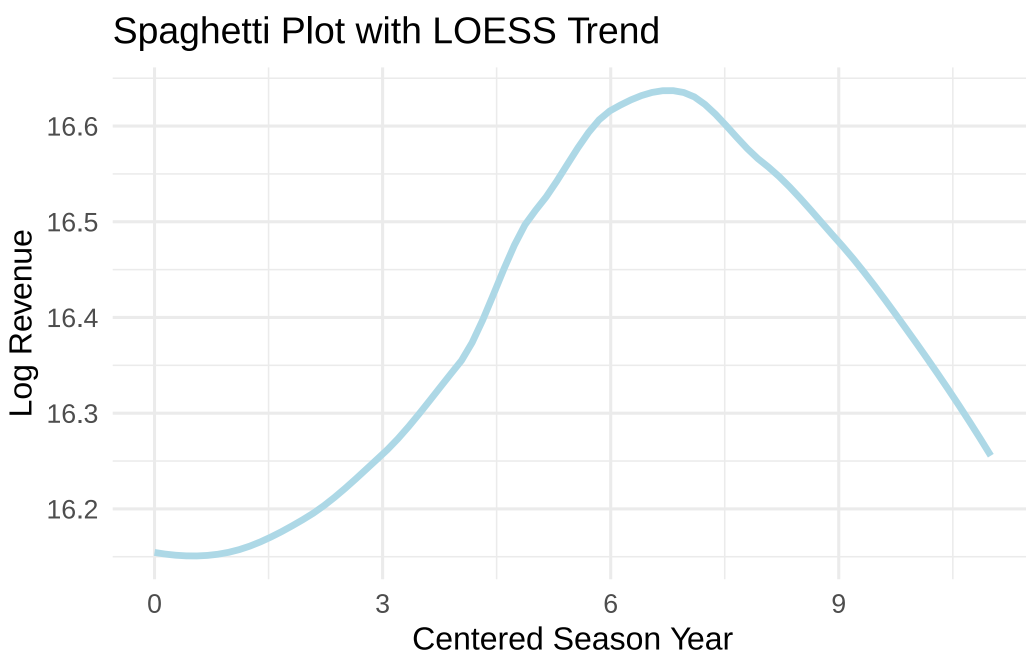
```
ggplot(clean_player_df) +
  geom_line(aes(x = season_year_c, y = usd_revenue_2024_log,
                group = player_id),color = "black", alpha = 0.15)
```




```r
ggplot(clean_player_df) +
  geom_line(aes(x = season_year_c, y = usd_revenue_2024_log,
            group = club_id,color = factor(clean_player_df$club_id)),
            alpha = 0.5)
```




```
ggplot(clean_player_df, aes(x = season_year_c,
                            y = usd_revenue_2024_log)) +

  # LOESS smoother (overall trend)
  geom_smooth(method = "loess",
              color = "lightblue", size = 1.2, se = FALSE) +
  labs(x = "Centered Season Year",
       y = "Log Revenue",
       title = "Spaghetti Plot with LOESS Trend") +
  theme_minimal(base_size = 12)
```



Modeling Process Plan

- Model 1: Random Intercepts only to examine ICC
- Testing whether a random slope on year is needed.
  - Model 2.1: Random intercepts + fixed effect of centered season_year
  - Model 2.2: Adding quadratic terms for season year
  - Model 3.3: Adding cubic term for season year
  - Model 2.4 Adding random slope of season_year_c
  - Model 2.5: Adding quadratic term of season_year as a random slope
- Adding level fixed effects
  - Establish whether age quadratic term is needed
    - Model 3.1: Adding player covariates {Height + Position + Citizen Status + Region + Age + Goals + Assists + Red Cards + Yellow cards + Minutes played}

        * Model 3.2: Adding the quadratic term for age

    - Model 4: Adding club covariates {UEFA coefficients + Foreign Players}

- Compare model fit
- Model diagnostic
- Choose final model

Model 1, intercepts only

```
#Using bobyqa to handle the complex random effects structure and avoid gradient warnings
strict_control <- lmerControl(optimizer = "bobyqa",
                              optCtrl = list(maxfun = 200000))

Model_1 <-  lmer(
  usd_revenue_2024_log ~
    # random effects
      (1|player_id) + (1|club_id), ,
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(digits = 4,  Model_1)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20105.1893
BIC	20132.6171
Pseudo-R^2 (fixed effects)	0.0000
Pseudo-R^2 (total)	0.7799

- We can see that player_id accounts for about 76% of market value while club_id accounts for 3%.

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.8205	0.0691	228.8952	26.9891	0.0000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.3209
club_id	(Intercept)	0.2837
Residual		0.7176

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7455
club_id	21	0.0344

Model 2.1: Adding fixed effect of year

```r
Model_2.1 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c +
    # random effects
      (1|player_id) + (1|club_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(digits = 4,  Model_2.1)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20052.8025
BIC	20087.0872
Pseudo-R^2 (fixed effects)	0.0048
Pseudo-R^2 (total)	0.7907

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.9908	0.0723	221.3121	32.9064	0.0000
season_year_c	-0.0310	0.0038	-8.1131	6524.0329	0.0000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.3441
club_id	(Intercept)	0.2822
Residual		0.7088

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7563
club_id	21	0.0333

Model 2.2 Adding quadratic term

- Notice that the linear relationship of year is no longer significant.

```
Model_2.2 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 2, raw = TRUE) +
    # random effects
      (1|player_id) + (1|club_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(digits = 4,  Model_2.2)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20046.9776
BIC	20088.1192
Pseudo-R^2 (fixed effects)	0.0058
Pseudo-R^2 (total)	0.7913

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.9164	0.0741	214.8843	36.5518	0.0000
poly(season_year_c, 2, raw = TRUE)1	0.0187	0.0118	1.5836	5756.8595	0.1133
poly(season_year_c, 2, raw = TRUE)2	-0.0045	0.0010	-4.4510	5650.5050	0.0000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.3434
club_id	(Intercept)	0.2818
Residual		0.7076

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7568
club_id	21	0.0333

Compare Model 2.1 and 2.2 to determine if the quadratic term is needed.

- We see that the LRT test suggests that the full model provides a significantly better fit than the reduce model, the quadratic term is needed for this data.

```
anova(Model_2.1, Model_2.2, test = "LRT") |> gt()
```

npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
5	20039.98	20074.27	-10014.99	20029.98	NA	NA	NA
6	20022.19	20063.33	-10005.10	20010.19	19.78947	1	0.000008645778

Model 2.3 Adding cubic term

- Now linear year trends is significant

```r
Model_2.3 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3, raw = TRUE) +
    # random effects
      (1|player_id) + (1|club_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(digits = 4,  Model_2.3)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20037.4036
BIC	20085.4022
Pseudo-R^2 (fixed effects)	0.0067
Pseudo-R^2 (total)	0.7930

Compare Model 2.2 and 2.3 to determine if the quadratic term is needed.

- We see that the LRT test suggests that the full model provides a significantly better fit than the reduce model, the cubic term is needed for this data. After adding cubic terms,

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.9932	0.0756	211.5628	39.5941	0.0000
poly(season_year_c, 3, raw = TRUE)1	-0.0926	0.0248	-3.7333	5460.0957	0.0002
poly(season_year_c, 3, raw = TRUE)2	0.0220	0.0053	4.1535	5343.5767	0.0000
poly(season_year_c, 3, raw = TRUE)3	-0.0016	0.0003	-5.0949	5343.9721	0.0000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.3458
club_id	(Intercept)	0.2819
Residual		0.7055

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7583
club_id	21	0.0333

it yields a better fit. It cannot only explain the peak of the overall trajectory, but can also explain the up and down wave pattern over time of the trajectory.

```r
anova(Model_2.2, Model_2.3, test = "LRT") |> gt()
```

Model 2.4, Random slope

- To estimate a random slope `(1 + year_centered | player_id)`, the model needs multiple data points for *every* player to calculate their individual trajectory. But many players only appear in our data for 1 or 2 years, and the model struggles to distinguish the "player slope" from the "residual error," causing these convergence issues. Clubs, however, have many players and many years of historical market value data so a random slope for clubs is more robust to fit.

npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
6	20022.19	20063.33	-10005.096	20010.19	NA	NA	NA
7	19998.32	20046.32	-9992.159	19984.32	25.87479	1	0.0000003642961

```
Model_2.4 <-  lmer(
  usd_revenue_2024_log ~
    ## first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3, raw = TRUE) +
    # random effects
      (1 | player_id) + (season_year_c | club_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(digits = 4,  Model_2.4)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	19974.7175
BIC	20036.4300

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.9823	0.0887	180.2113	28.0241	0.0000
poly(season_year_c, 3, raw = TRUE)1	-0.0920	0.0260	-3.5333	1018.9716	0.0004
poly(season_year_c, 3, raw = TRUE)2	0.0226	0.0052	4.3039	5311.3234	0.0000
poly(season_year_c, 3, raw = TRUE)3	-0.0016	0.0003	-5.2901	5311.7328	0.0000

p values calculated using Satterthwaite d.f.

Model 2.4, Random slope + quadratic term

- We have a convergence problem, it seems that adding the quadratic term makes the model unidentifiable. We move forward with Models 2.2 and 2.3.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.3508
club_id	(Intercept)	0.3532
club_id	season_year_c	0.0394
Residual		0.6975

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7491
club_id	21	0.0512

```
Model_2.5 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3, raw = TRUE) +
    # random effects
      (1 | player_id) + (season_year_c + season_year_c^2 | club_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(digits = 4,  Model_2.5)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	19974.7175
BIC	20036.4300

Determine if the random slope on season year is needed

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.9823	0.0887	180.2113	28.0241	0.0000
poly(season_year_c, 3, raw = TRUE)1	-0.0920	0.0260	-3.5333	1018.9716	0.0004
poly(season_year_c, 3, raw = TRUE)2	0.0226	0.0052	4.3039	5311.3234	0.0000
poly(season_year_c, 3, raw = TRUE)3	-0.0016	0.0003	-5.2901	5311.7328	0.0000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.3508
club_id	(Intercept)	0.3532
club_id	season_year_c	0.0394
Residual		0.6975

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7491
club_id	21	0.0512

```r
calculate_lrt_mixture <- function(reduced_model, full_model) {

  # 1. Extract Log-Likelihoods
  ll_reduced <- as.numeric(logLik(reduced_model))
  ll_full    <- as.numeric(logLik(full_model))

  # 2. Calculate LRT Statistic
  # Formula: -2 * (LL_reduced - LL_full)
  lrt_stat <- -2 * (ll_reduced - ll_full)

  # 3. Calculate Difference in Parameters (df)
  # This tells us how many params we added
  df_reduced <- attr(logLik(reduced_model), "df")
  df_full    <- attr(logLik(full_model), "df")
  df_diff    <- df_full - df_reduced

  # 4. Calculate P-Value based on the difference
```

```r
  # If we added a slope + covariance, df_diff should be 2.
  # The mixture is 0.5 * ChiSq(1) + 0.5 * ChiSq(2)

  if (df_diff == 2) {
    p_val <- 0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE) +
             0.5 * pchisq(lrt_stat, df = 2, lower.tail = FALSE)

    msg <- "Mixture of ChiSq(1) and ChiSq(2)"

  } else if (df_diff == 1) {
    # If we added a parameter with NO covariance (e.g. diag structure)
    p_val <- 0.5 * pchisq(lrt_stat, df = 0, lower.tail = FALSE) +
             0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE)

    msg <- "Mixture of ChiSq(0) and ChiSq(1)"

  } else {
    # Fallback for non-standard comparisons
    p_val <- pchisq(lrt_stat, df = df_diff, lower.tail = FALSE)
    msg <- paste0("Standard ChiSq(", df_diff, ")")
  }

  # Output
  cat("LRT Statistic:", round(lrt_stat, 4), "\n")
  cat("Parameter Diff:", df_diff, "\n")
  cat("Distribution:  ", msg, "\n")
  cat("P-Value:       ", format.pval(p_val, eps = .0001), "\n")
}

# Run the comparison
calculate_lrt_mixture(Model_2.3, Model_2.4)
```

```
LRT Statistic: 66.6861
Parameter Diff: 2
Distribution:   Mixture of ChiSq(1) and ChiSq(2)
P-Value:        < 0.0001
```

- We use a helper function to print the p-value for the LRT difference, and find a p-value $<.0001$ and we reject the Null, (with $.5 \times \chi^2(1) + .5 \times \chi^2(2)$ mixture) and conclude that there is strong evidence for club variation in market value trajectories and therefore we should keep this random slope in the model.

Model 3.1: Adding fixed effects at level 1 with age as a linear relationship

- Age centered is not even significant. Same for red\yellow cards.

```
Model_3.1 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3, raw = TRUE) +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    age_c +
    goals + assists + red_cards + yellow_cards + minutes_played +
    # random effects
    (1 | player_id) + (season_year_c | club_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(digits = 4,  Model_3.1)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	17469.2423
BIC	17640.6660

Model 3.2. Adding age quadratic term

```
Model_3.2 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3, raw = TRUE) +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    poly(age_c, 2, raw = TRUE) +
    goals + assists + red_cards + yellow_cards + minutes_played +
```

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	16.5121	0.1020	161.8997	44.4734	0.0000
poly(season_year_c, 3, raw = TRUE)1	-0.0538	0.0226	-2.3814	1148.4380	0.0174
poly(season_year_c, 3, raw = TRUE)2	0.0222	0.0046	4.8478	5237.4683	0.0000
poly(season_year_c, 3, raw = TRUE)3	-0.0018	0.0003	-6.6889	5239.0873	0.0000
height_c	0.0140	0.0037	3.7216	1669.4917	0.0002
positionDefender	-0.4310	0.0596	-7.2306	1650.1235	0.0000
positionGoalkeeper	-1.2404	0.0976	-12.7037	1666.8564	0.0000
positionMidfield	-0.2276	0.0595	-3.8227	1677.8494	0.0001
citizen_of_club_countryYes	-0.3992	0.0376	-10.6299	5729.2044	0.0000
RegionEast Asia & Pacific	-0.1817	0.2112	-0.8605	1734.4279	0.3896
RegionLatin America & Caribbean	0.2481	0.0634	3.9134	1796.5122	0.0001
RegionMiddle East & North Africa	0.0833	0.2452	0.3396	1697.4147	0.7342
RegionNorth America	-0.0053	0.2249	-0.0234	1778.5037	0.9814
RegionSub-Saharan Africa	0.0183	0.1038	0.1759	1732.4873	0.8604
age_c	-0.0010	0.0040	-0.2525	2354.7908	0.8007
goals	0.0139	0.0026	5.3168	5538.6861	0.0000
assists	0.0126	0.0036	3.4445	5075.1564	0.0006
red_cards	-0.0187	0.0296	-0.6338	4696.8271	0.5262
yellow_cards	0.0058	0.0041	1.4203	5406.1624	0.1556
minutes_played	0.5115	0.0174	29.4113	5605.9443	0.0000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.9464
club_id	(Intercept)	0.3686
club_id	season_year_c	0.0329
Residual		0.6145

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.6356
club_id	21	0.0964

```r
    # random effects
    (1 | player_id) + (season_year_c | club_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(digits = 4,  Model_3.2)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	14507.0975
BIC	14685.3781

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	16.8498	0.0783	215.3074	82.3319	0.0000
poly(season_year_c, 3, raw = TRUE)1	-0.0557	0.0169	-3.2907	2124.8495	0.0010
poly(season_year_c, 3, raw = TRUE)2	0.0276	0.0035	7.9468	5036.1261	0.0000
poly(season_year_c, 3, raw = TRUE)3	-0.0022	0.0002	-10.8944	5037.4463	0.0000
height_c	0.0098	0.0036	2.6949	1799.8481	0.0071
positionDefender	-0.4037	0.0578	-6.9786	1786.1940	0.0000
positionGoalkeeper	-1.0571	0.0947	-11.1594	1797.7097	0.0000
positionMidfield	-0.1905	0.0577	-3.3014	1802.0409	0.0010
citizen_of_club_countryYes	-0.1898	0.0314	-6.0394	6875.7653	0.0000
RegionEast Asia & Pacific	-0.4305	0.2041	-2.1098	1836.2341	0.0350
RegionLatin America & Caribbean	0.1908	0.0609	3.1332	1954.2127	0.0018
RegionMiddle East & North Africa	-0.0917	0.2374	-0.3864	1814.6163	0.6992
RegionNorth America	-0.0634	0.2169	-0.2923	1856.8054	0.7701
RegionSub-Saharan Africa	0.0659	0.1003	0.6570	1848.9110	0.5113
poly(age_c, 2, raw = TRUE)1	0.0186	0.0037	4.9884	2295.0021	0.0000
poly(age_c, 2, raw = TRUE)2	-0.0242	0.0004	-62.7957	6152.2162	0.0000
goals	0.0047	0.0020	2.3634	5169.1922	0.0181
assists	0.0095	0.0028	3.4497	4882.6031	0.0006
red_cards	0.0095	0.0222	0.4279	4678.4516	0.6687
yellow_cards	-0.0018	0.0031	-0.5826	5089.4375	0.5602
minutes_played	0.3528	0.0134	26.2596	5243.8360	0.0000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.9599
club_id	(Intercept)	0.2373
club_id	season_year_c	0.0186
Residual		0.4589

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7754
club_id	21	0.0474

- It seems that we need the quadratic term for age in the model.

```
anova(Model_3.1, Model_3.2) |> gt()
```

Model 4: Adding club covariates

- The model with club covariates under performs compared to model 3.2, yet foreign players is significant. We generate a new model 4 dropping UEFA coefficients and test against 3.2 again.

```
Model_4 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3, raw = TRUE) +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    poly(age_c, 2, raw = TRUE) +
    goals + assists + red_cards + yellow_cards + minutes_played +
    # club fixed effects
```

npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
25	17345.33	17516.75	-8647.664	17295.33	NA	NA	NA
26	14362.95	14541.23	-7155.476	14310.95	2984.376	1	0

```r
    uefa_coefficient + foreign_players +
    # random effects
    (1 | player_id) + (season_year_c | club_id),
  data = clean_player_df, REML = TRUE,    control = strict_control)

summ(digits = 4,  Model_4)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	14520.4840
BIC	14712.4785

```r
anova(Model_3.2, Model_4) |> gt()
```

- This time, model 4 is a better fit with only 1 club covariate. AIC is lower for Model 4, but BIC is higher for Model 3.2. We decide to keep Model 4 as it contains club contextual covariates information.

```r
Model_4 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3, raw = TRUE) +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    poly(age_c, 2, raw = TRUE) +
    goals + assists + red_cards + yellow_cards + minutes_played +
    # club fixed effects
    foreign_players +
    # random effects
    (1 | player_id) + (season_year_c | club_id),
  data = clean_player_df, REML = TRUE,    control = strict_control)
```

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	16.8574	0.0793	212.4550	77.4322	0.0000
poly(season_year_c, 3, raw = TRUE)1	-0.0610	0.0172	-3.5499	1979.2430	0.0004
poly(season_year_c, 3, raw = TRUE)2	0.0283	0.0035	8.0967	5037.1907	0.0000
poly(season_year_c, 3, raw = TRUE)3	-0.0023	0.0002	-10.9692	5035.6950	0.0000
height_c	0.0098	0.0036	2.7041	1797.2098	0.0069
positionDefender	-0.4040	0.0578	-6.9914	1783.3530	0.0000
positionGoalkeeper	-1.0566	0.0946	-11.1657	1795.0980	0.0000
positionMidfield	-0.1903	0.0576	-3.3020	1799.2532	0.0010
citizen_of_club_countryYes	-0.1860	0.0315	-5.9054	6884.1551	0.0000
RegionEast Asia & Pacific	-0.4307	0.2039	-2.1129	1833.7205	0.0347
RegionLatin America & Caribbean	0.1889	0.0608	3.1056	1951.9683	0.0019
RegionMiddle East & North Africa	-0.0936	0.2372	-0.3948	1811.8833	0.6930
RegionNorth America	-0.0659	0.2167	-0.3040	1854.3493	0.7612
RegionSub-Saharan Africa	0.0638	0.1002	0.6369	1846.5714	0.5243
poly(age_c, 2, raw = TRUE)1	0.0184	0.0037	4.9545	2294.6335	0.0000
poly(age_c, 2, raw = TRUE)2	-0.0242	0.0004	-62.7901	6163.0575	0.0000
goals	0.0048	0.0020	2.3857	5164.5854	0.0171
assists	0.0096	0.0028	3.4731	4878.2160	0.0005
red_cards	0.0087	0.0223	0.3907	4673.6600	0.6960
yellow_cards	-0.0018	0.0031	-0.5724	5084.8163	0.5671
minutes_played	0.3529	0.0134	26.2699	5239.3341	0.0000
uefa_coefficient	-0.0011	0.0012	-0.9810	4466.8449	0.3267
foreign_players	0.1995	0.0971	2.0533	1972.8895	0.0402

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.9588
club_id	(Intercept)	0.2445
club_id	season_year_c	0.0194
Residual		0.4589

```
summ(digits = 4,  Model_4)
```

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7728
club_id	21	0.0502

npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
26	14362.95	14541.23	-7155.476	14310.95	NA	NA	NA
28	14361.89	14553.88	-7152.944	14305.89	5.064713	2	0.07947152

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	14507.7590
BIC	14692.8966

```
anova(Model_3.2, Model_4) |> gt()
```

Bonus round, Model 5 testing 2 levels only for player data.

- We drop the random intercept for club and move the random slope of season year to player and find that this is a worse fit than Model 4.
- Model 4 is our final model

```
Model_5 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3, raw = TRUE) +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    poly(age_c, 2, raw = TRUE) +
    goals + assists + red_cards + yellow_cards + minutes_played +
    # club fixed effects
    foreign_players +
```

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	16.8572	0.0790	213.4409	79.1128	0.0000
poly(season_year_c, 3, raw = TRUE)1	-0.0611	0.0172	-3.5602	2024.2364	0.0004
poly(season_year_c, 3, raw = TRUE)2	0.0283	0.0035	8.1128	5037.2537	0.0000
poly(season_year_c, 3, raw = TRUE)3	-0.0023	0.0002	-10.9890	5035.1017	0.0000
height_c	0.0098	0.0036	2.6981	1796.9647	0.0070
positionDefender	-0.4040	0.0578	-6.9923	1783.2364	0.0000
positionGoalkeeper	-1.0573	0.0946	-11.1737	1794.7986	0.0000
positionMidfield	-0.1905	0.0576	-3.3047	1799.1086	0.0010
citizen_of_club_countryYes	-0.1861	0.0315	-5.9085	6884.2560	0.0000
RegionEast Asia & Pacific	-0.4316	0.2038	-2.1172	1833.5081	0.0344
RegionLatin America & Caribbean	0.1889	0.0608	3.1055	1951.8629	0.0019
RegionMiddle East & North Africa	-0.0939	0.2372	-0.3958	1811.7669	0.6923
RegionNorth America	-0.0666	0.2166	-0.3073	1854.1815	0.7586
RegionSub-Saharan Africa	0.0635	0.1002	0.6339	1846.4296	0.5262
poly(age_c, 2, raw = TRUE)1	0.0184	0.0037	4.9562	2294.5304	0.0000
poly(age_c, 2, raw = TRUE)2	-0.0242	0.0004	-62.7837	6159.9545	0.0000
goals	0.0047	0.0020	2.3749	5166.6298	0.0176
assists	0.0096	0.0028	3.4706	4879.7043	0.0005
red_cards	0.0088	0.0223	0.3960	4674.9908	0.6921
yellow_cards	-0.0018	0.0031	-0.5770	5086.4328	0.5640
minutes_played	0.3530	0.0134	26.2763	5241.1391	0.0000
foreign_players	0.1994	0.0971	2.0535	1948.7098	0.0402

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.9587
club_id	(Intercept)	0.2420
club_id	season_year_c	0.0192
Residual		0.4589

```
    # random effects
    (season_year_c | player_id),
  data = clean_player_df, REML = TRUE,    control = strict_control)
```

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7735
club_id	21	0.0493

npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
26	14362.95	14541.23	-7155.476	14310.95	NA	NA	NA
27	14360.80	14545.94	-7153.401	14306.80	4.150156	1	0.04163068

```r
summ(digits = 4,  Model_5)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	14002.8243
BIC	14181.1049

```r
anova(Model_4, Model_5) |> gt()
```

```r
export_summs(Model_2.3, Model_3.2, Model_4,
              digits=3, r.squared=FALSE,
             model.names = c("Model_2.3", "Model_3.2", "Model_4"))
```

```r
final_model <- Model_4
```

Model Diagnostics

- By comparing the QQ-line and QQ-plot (both residuals and random effects) we can find that both model plots look good on normality, except some outliers. By comparing the residual-fitted value plot, we can find that both model show the same spread pattern, the dots are evenly spread across zero-mean line up and down.

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	16.8680	0.0689	244.8751	1809.6821	0.0000
poly(season_year_c, 3, raw = TRUE)1	-0.0464	0.0160	-2.9063	5578.4734	0.0037
poly(season_year_c, 3, raw = TRUE)2	0.0283	0.0032	8.8677	4869.2439	0.0000
poly(season_year_c, 3, raw = TRUE)3	-0.0024	0.0002	-12.9775	4735.8699	0.0000
height_c	0.0091	0.0041	2.2023	1793.9566	0.0278
positionDefender	-0.3925	0.0661	-5.9398	1798.9855	0.0000
positionGoalkeeper	-1.0698	0.1083	-9.8743	1818.2584	0.0000
positionMidfield	-0.1723	0.0660	-2.6118	1810.7953	0.0091
citizen_of_club_countryYes	-0.1747	0.0316	-5.5340	6799.0627	0.0000
RegionEast Asia & Pacific	-0.5234	0.2313	-2.2626	1781.2416	0.0238
RegionLatin America & Caribbean	0.0771	0.0681	1.1329	1838.8817	0.2574
RegionMiddle East & North Africa	-0.2125	0.2768	-0.7676	1868.5437	0.4428
RegionNorth America	-0.1882	0.2494	-0.7544	1927.8988	0.4507
RegionSub-Saharan Africa	0.0501	0.1137	0.4409	1832.2880	0.6594
poly(age_c, 2, raw = TRUE)1	0.0291	0.0045	6.4544	2280.5012	0.0000
poly(age_c, 2, raw = TRUE)2	-0.0287	0.0005	-53.7915	3158.2327	0.0000
goals	0.0036	0.0018	2.0041	4107.8687	0.0451
assists	0.0041	0.0024	1.6894	3817.3205	0.0912
red_cards	-0.0059	0.0189	-0.3108	3642.5735	0.7560
yellow_cards	-0.0048	0.0027	-1.7629	3935.1087	0.0780
minutes_played	0.2688	0.0119	22.5382	4240.1634	0.0000
foreign_players	0.1746	0.0734	2.3777	4890.8417	0.0175

p values calculated using Satterthwaite d.f.

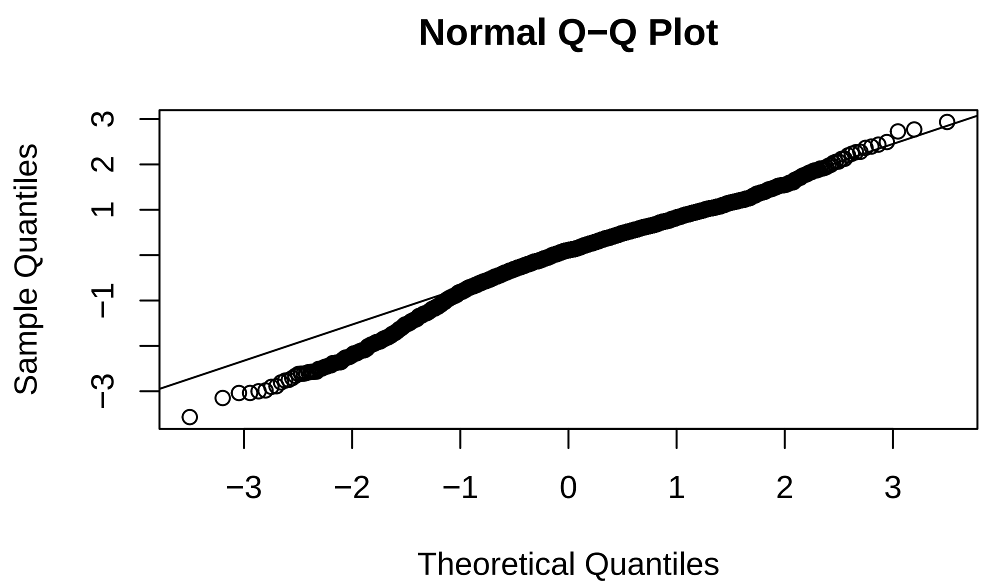
Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.2404
player_id	season_year_c	0.1395
Residual		0.3639

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.9207

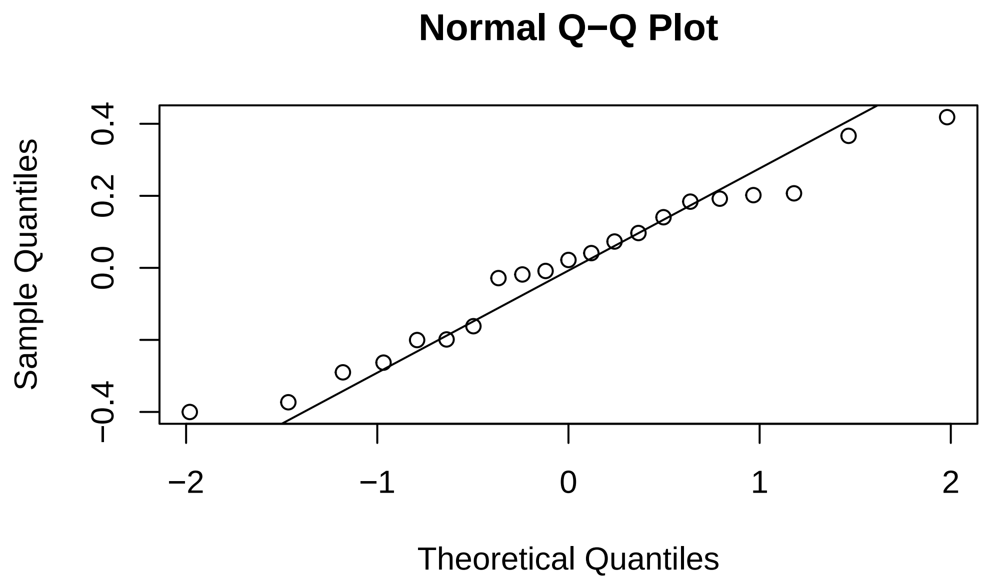
npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
26	13855.91	14034.19	-6901.956	13803.91	NA	NA	NA
27	14360.80	14545.94	-7153.401	14306.80	0	1	1

- By comparing EBLUPs plots, we can see that most of the points lie deviate from the diagonal at both tails, and the lower side is more downward, suggesting moderate departures from normality in the random slopes and random intercepts, which indicates possible skewness and outliers.Thus, we cannot say the normal random effects assumption in the multilevel model are met.

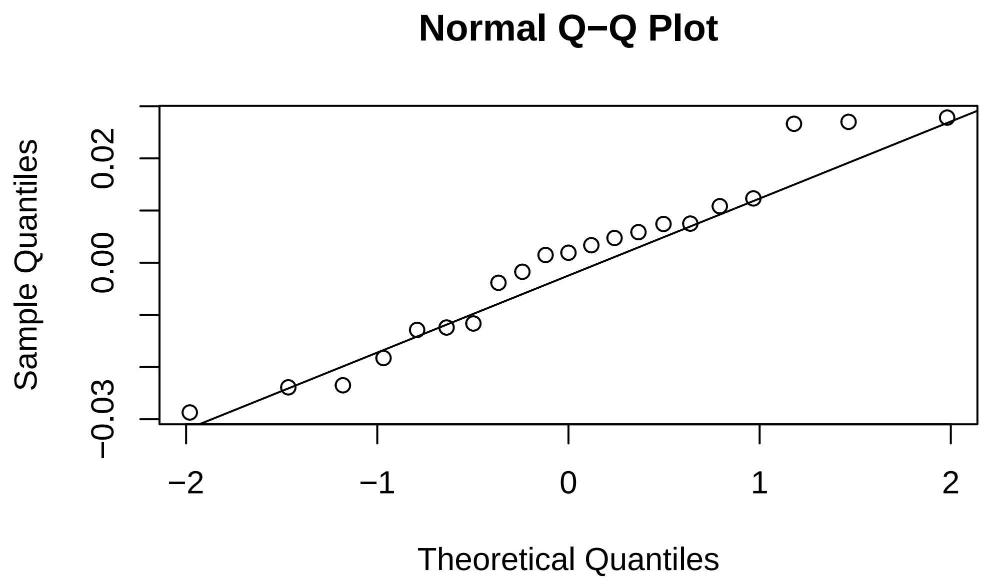
```
# slopes
qqnorm(ranef(final_model)$player_id[,1]) # group factor
qqline(ranef(final_model)$player_id[,1])
```




```
qqnorm(ranef(final_model)$club_id[,1]) # group factor
qqline(ranef(final_model)$club_id[,1])
```

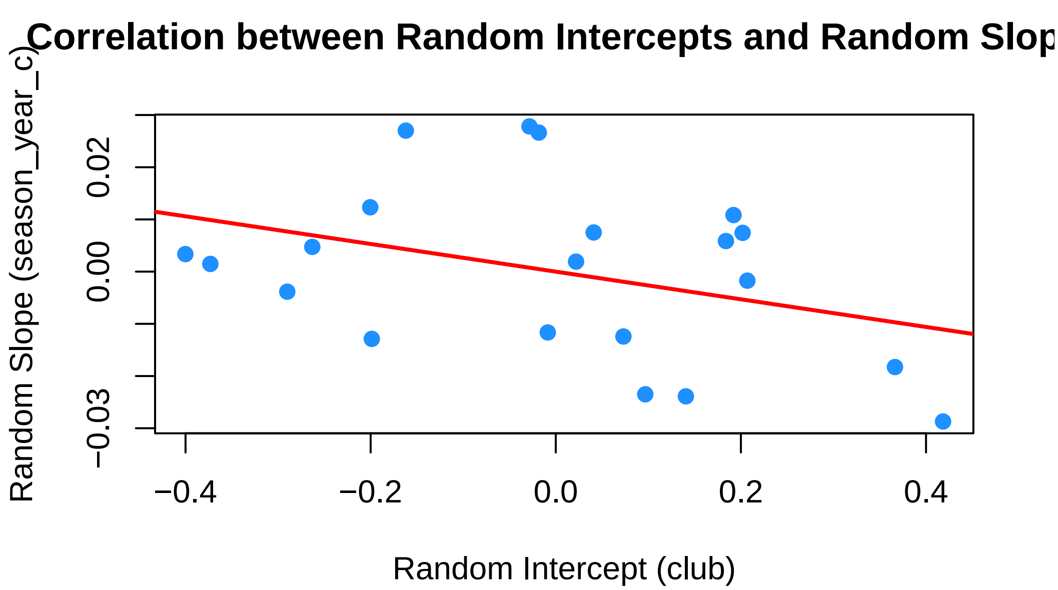



```r
qqnorm(ranef(final_model)$club_id[,2]) # group factor
qqline(ranef(final_model)$club_id[,2])
```

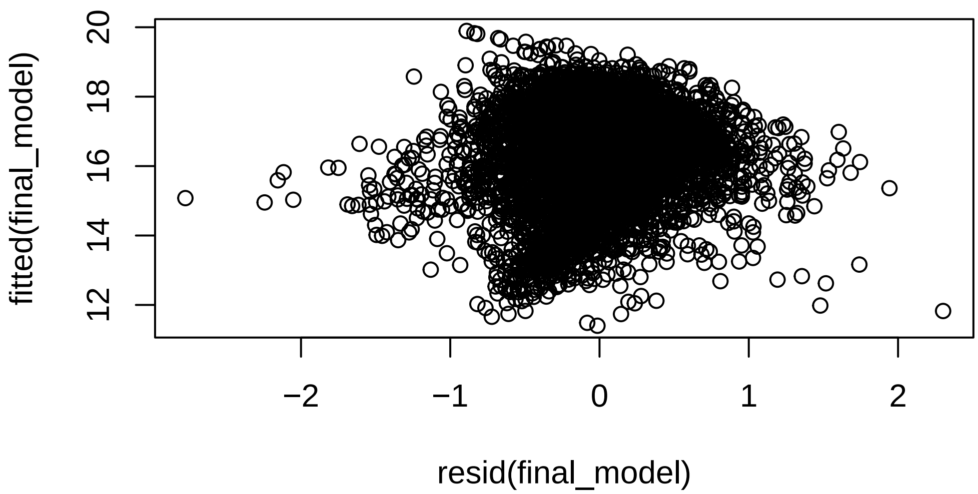


- The red regression line slopes slightly downward, but only gently. That is, clubs with higher baseline log market value tend to have slightly more negative or flatter slopes. Clubs with lower baseline market value tend to have slightly steeper slopes.
- Most residuals lie between about −1 and +1, and fitted log revenues between 14 and 18. There is no strong curved pattern or obvious systematic relationship between residual and fitted values. So the residual reasonably consistent with the linearity assumption. Besides, there is no funnel shape, so the residuals are homogeneous. However, we can still see some outliers exist.

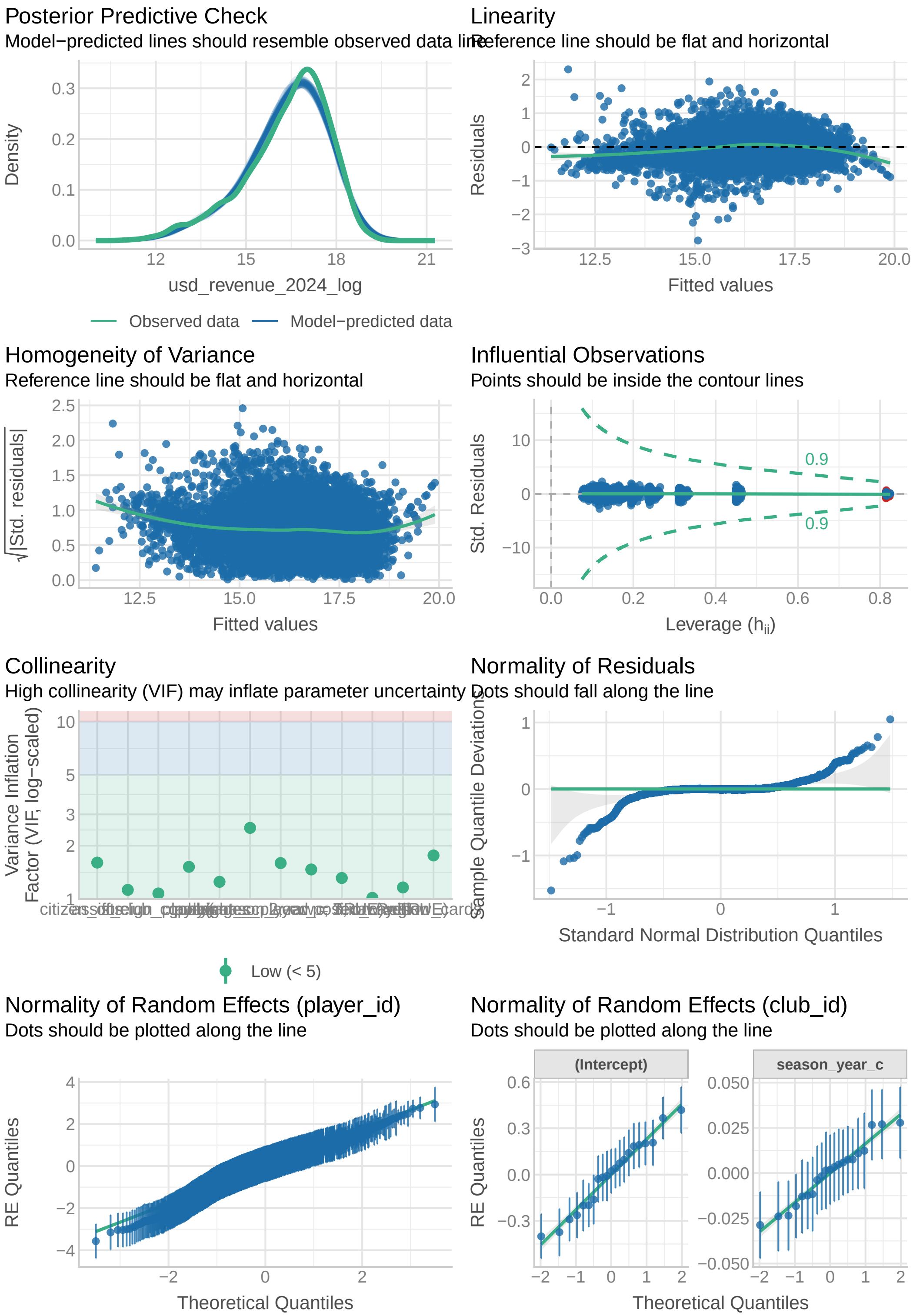
```r
re <- ranef(final_model)$club_id

plot(
  re[, "(Intercept)"],
  re[, "season_year_c"],
  xlab = "Random Intercept (club)",
  ylab = "Random Slope (season_year_c)",
  main = "Correlation between Random Intercepts and Random Slopes",
  pch = 19, col = "dodgerblue"
)

abline(lm(re[, "season_year_c"] ~ re[, "(Intercept)"]), col = "red", lwd = 2)
```



```r
# Residual scatterplot
plot(resid(final_model),fitted(final_model))
```




```r
check_model(final_model)
```


Posterior Predictive Check
Model−predicted lines should resemble observed data line
Density
usd_revenue_2024_log
Observed data
Model−predicted data
Linearity
Reference line should be flat and horizontal
Residuals
Fitted values
Homogeneity of Variance
Reference line should be flat and horizontal
√|Std. residuals|
Fitted values
Influential Observations
Points should be inside the contour lines
Std. Residuals
Leverage (h_{ii})
0.9
Collinearity
High collinearity (VIF) may inflate parameter uncertainty
Variance Inflation Factor (VIF, log−scaled)
Low (< 5)
Normality of Residuals
Dots should fall along the line
Sample Quantile Deviations
Standard Normal Distribution Quantiles
Normality of Random Effects (player_id)
Dots should be plotted along the line
RE Quantiles
Theoretical Quantiles
Normality of Random Effects (club_id)
Dots should be plotted along the line
(Intercept)
season_year_c
RE Quantiles
Theoretical Quantiles

Final Model Specification

To test our hypotheses regarding the determinants of player market value, we fitted a **cross-classified multilevel model** (Model 4). This structure accounts for the non-hierarchical nature of the data, where players (i) move between different clubs (j) over time (t).

The final model specification for the log-transformed market value, Y_{tij}, is formally expressed as:

$$
\begin{aligned}
\log(\text{Revenue})_{tij} = \beta_0 &+ \underbrace{\beta_1(\text{Year}_t) + \beta_2(\text{Year}_t^2) + \beta_3(\text{Year}_t^3)}_{\text{Cubic Time Trends}} \\
&+ \underbrace{\beta_4(\text{Height}_i - \bar{H}) + \beta_5(\text{Citizen}_{ij}) + \beta_6(\text{Age}_{it} - \bar{A}) + \beta_7(\text{Age}_{it}^2)}_{\text{Player Fixed Effects}} \\
&+ \underbrace{\sum_p \beta_{Pos,p}(\mathbb{I}_{\text{Position}_i = p}) + \sum_r \beta_{Reg,r}(\mathbb{I}_{\text{Region}_i = r})}_{\text{Categorical Controls}} \\
&+ \underbrace{\beta_8(\text{Goals}_{it}^*) + \beta_9(\text{Assists}_{it}^*) + \beta_{10}\text{Red}_{it} + \beta_{11}\text{Yellow}_{it} + \beta_{12}(\text{Mins}_{Z,it})}_{\text{Performance \& Status}} \\
&+ \underbrace{\beta_{13}(\text{Foreign}_{tj} - \bar{F})}_{\text{Club Fixed Effect}} \\
&+ u_{0i} + v_{0j} + v_{1j}(\text{Year}_t) + \epsilon_{tij}
\end{aligned}
$$

Interpretation of Results

- **Market Trajectory (Cubic Trend):** The model identifies a complex, non-linear market trajectory with a significant cubic term ($\beta_3 \approx -0.002, p < .001$). This "S-shaped" curve suggests the market initially stagnated (2013-2014), accelerated rapidly during the "boom" years (2015-2019), and has recently plateaued in the post-2020 era.
- **Club Internationalization ($\beta_{13} = 0.20$):** We find a significant premium for club diversity. A **10 percentage point increase** in the proportion of foreign players is associated with approximately a **2% increase** in player value. This suggests that clubs with higher international representation—likely a proxy for global reach—elevate the valuation of their entire squad.
- **Player Status ($\beta_{12} = 0.35$):** The variable `minutes_played` was standardized (Z-scored). The positive coefficient indicates that a **one standard deviation increase** in playing time is associated with a **35% increase** in market value, confirming that "Squad Status" (Starter vs. Bench) is a primary driver of valuation.

- **Homegrown Effect ($\beta_5 = -0.19$):** Domestic players (`citizen` = Yes) are valued lower than their non-citizen counterparts. Converting from the log scale ($e^{-0.19} - 1 \approx -17\%$), we find that **domestic players are valued approximately 17% lower** than foreign players.

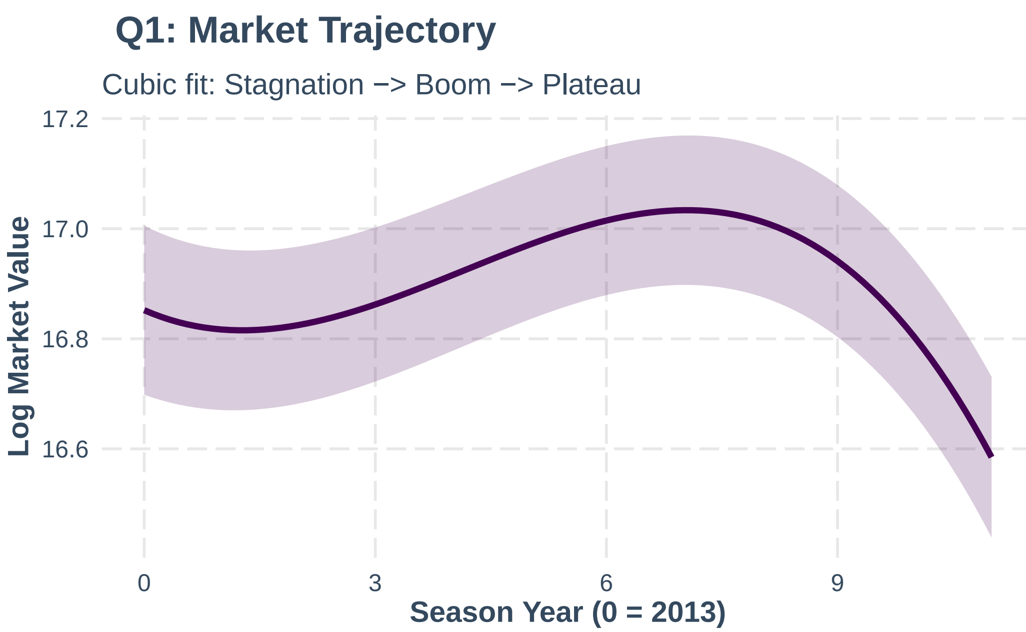
Visualization of Marginal Effects

Q1: Market Evolution

How does market value change annually compared to the 2013 baseline?

- The visualization answers Q1 by displaying a statistically significant S-shaped trajectory ($\beta_{cubic} \approx -0.002$). Rather than constant inflation, the market shows three distinct phases: an initial period of stability (2013-2014), a rapid "Boom" period of hyper-inflation (2015-2019), and a recent "Plateau" or deceleration in the post-2020 era.

```
effect_plot(final_model, pred = season_year_c, interval = TRUE,
            colors = "#440154", # Viridis Purple
            x.label = "Season Year (0 = 2013)",
            y.label = "Log Market Value") +
  labs(title = "Q1: Market Trajectory",
       subtitle = "Cubic fit: Stagnation -> Boom -> Plateau") +
  theme(plot.title = element_text(face = "bold"))
```

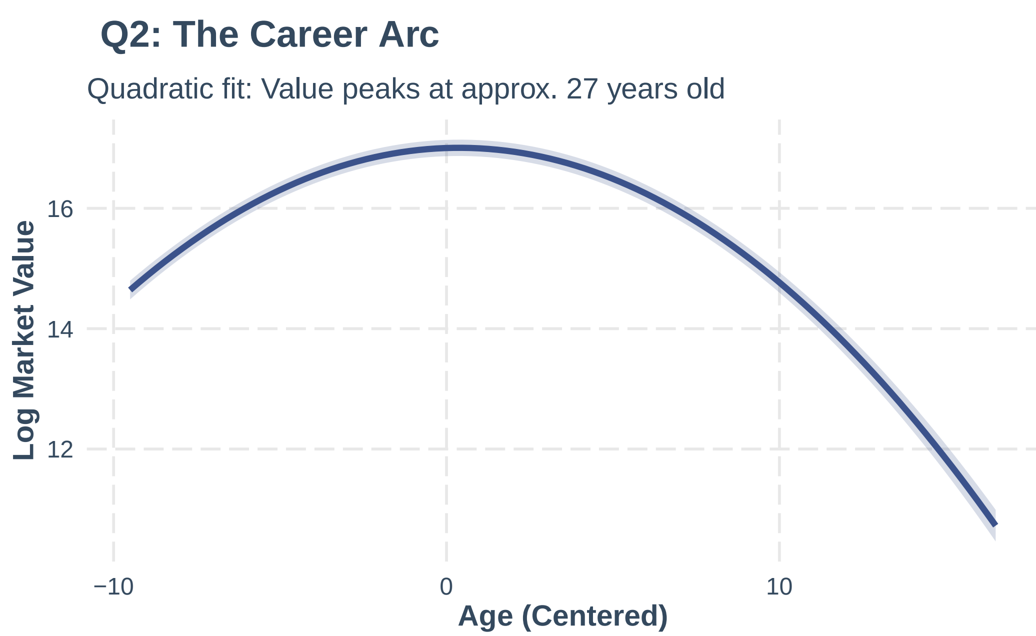


Q2: Biological Career Arc

How does market value change as a player moves through their biological career arc (Prospect $\rightarrow$ Prime $\rightarrow$ Veteran)?

- This plot answers Q2 by visualizing the significant quadratic effect of Age ($\beta_{Age^2} = -0.02$). The Inverted-U shape confirms that market value is biologically constrained: it rises steeply as players develop (Prospects), peaks during their physical prime (ages 26–28), and declines significantly as they become Veterans, reflecting the market's discount on older assets with lower resale potential.

```r
effect_plot(final_model, pred = age_c, interval = TRUE,
            colors = "#3b528b", # Viridis Blue
            x.label = "Age (Centered)",
            y.label = "Log Market Value") +
  labs(title = "Q2: The Career Arc",
       subtitle = "Quadratic fit: Value peaks at approx. 27 years old") +
  theme(plot.title = element_text(face = "bold"))
```

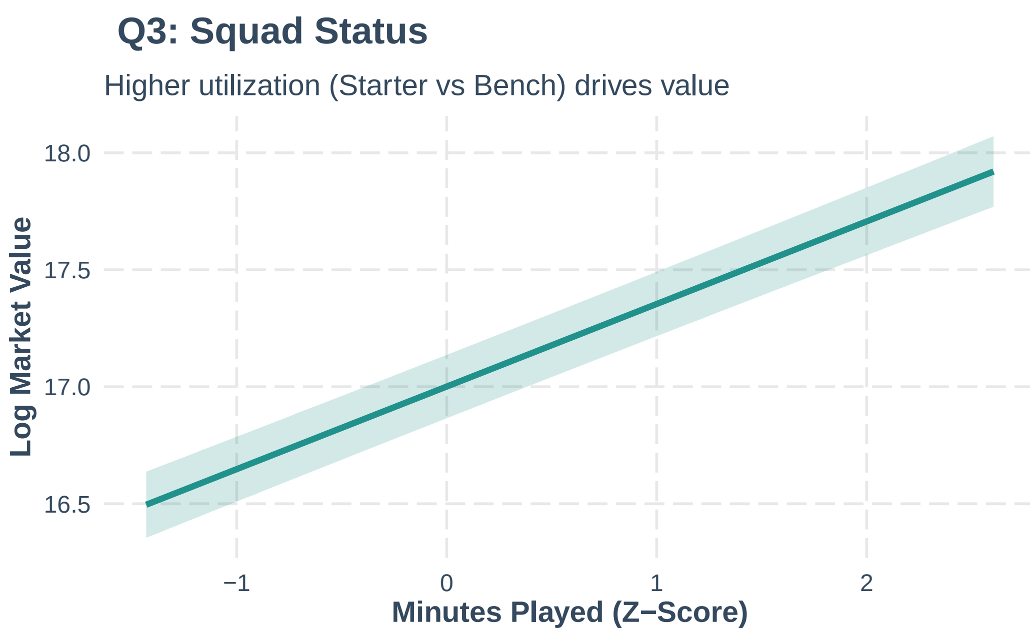


Q3: Squad Status

Does being a key player (playing more than the average teammate) increase market value?

- This visualization answers Q3 with a steep, positive linear slope ($\beta = 0.35$). Because minutes_played was standardized, the x-axis represents standard deviations. Moving from a "Bench Player" (-1 SD) to a "Key Starter" (+1 SD) results in a massive increase in valuation, confirming that Squad Status is one of the strongest predictors in the model.

```
effect_plot(final_model, pred = minutes_played, interval = TRUE,
            colors = "#21918c", # Viridis Teal
            x.label = "Minutes Played (Z-Score)",
            y.label = "Log Market Value") +
  labs(title = "Q3: Squad Status",
       subtitle = "Higher utilization (Starter vs Bench) drives value") +
  theme(plot.title = element_text(face = "bold"))
```

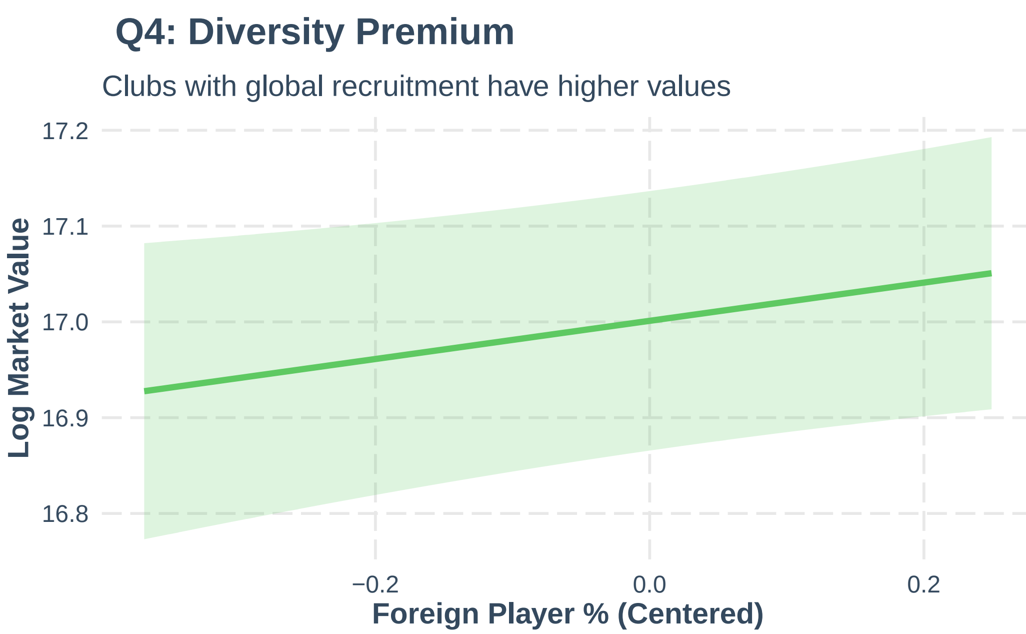


Q4: Club Diversity

Does having a higher proportion of foreign players than the average club increase market value?

- This plot answers Q4 by showing a positive relationship ($\beta = 0.20$) between club internationalization and player value. As a club's roster becomes more diverse (moving right on the x-axis), the predicted value of its players increases. This likely acts as a proxy for the financial power and global reach of the club.

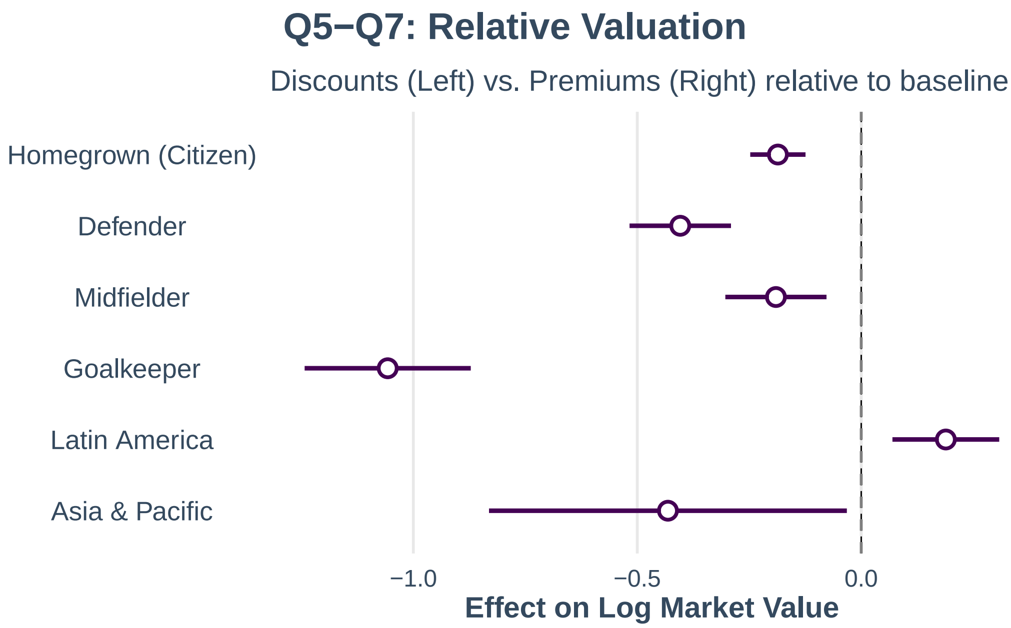
```r
effect_plot(final_model, pred = foreign_players, interval = TRUE,
            colors = "#5ec962", # Viridis Light Green
            x.label = "Foreign Player % (Centered)",
            y.label = "Log Market Value") +
  labs(title = "Q4: Diversity Premium",
       subtitle = "Clubs with global recruitment have higher values") +
  theme(plot.title = element_text(face = "bold"))
```



Q5, Q6, & Q7: Relative Valuation (Forest Plot)

- Q5 (Homegrown): Is there a 'homegrown premium' or discount?
- The negative coefficient for "Homegrown (Citizen)" confirms a Domestic Discount (or Foreign Premium). Local players are valued significantly lower than comparable foreign imports.
- Q6 (Position): How much less is a defender valued compared to an attacker?
- All defensive positions fall to the left of 0, confirming that Attackers (the baseline) hold the highest market value, with Goalkeepers facing the steepest discount.
- Q7 (Region): What is the market premium for players in Europe vs. others?
- While most regions overlap with 0 (no difference from Europe), Latin America shows a positive premium, suggesting players from this region act as a premium "export" brand in the global market.

```
plot_summs(final_model,
           coefs = c("Homegrown (Citizen)" = "citizen_of_club_countryYes",
                     "Defender" = "positionDefender",
                     "Midfielder" = "positionMidfield",
                     "Goalkeeper" = "positionGoalkeeper",
                     "Latin America" = "RegionLatin America & Caribbean",
                     "Asia & Pacific" = "RegionEast Asia & Pacific"),
           colors = "#440154") + # Viridis Purple
  labs(title = "Q5-Q7: Relative Valuation",
       subtitle = "Discounts (Left) vs. Premiums (Right) relative to baseline",
       x = "Effect on Log Market Value") +
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray50") +
  theme(plot.title = element_text(face = "bold"))
```



Limitations

While our model demonstrates strong predictive power (R2 0.89), several limitations warrant mention:

1. **Small Cluster Size for Random Slopes:** We estimated random slopes for `Club`, but our dataset contains only 21 unique clubs. While the Likelihood Ratio Test supported the inclusion of this slope ($p<.001$), variance estimates from small numbers of clusters should be interpreted with caution.
2. **Normality of Residuals:** Diagnostic plots (QQ plots) indicated deviation from normality in the extreme tails of the distribution. While the large sample size ($N=7023$) ensures robust estimation of fixed effects, future iterations could employ robust standard errors or a Gamma GLMM to better handle these extremes.
3. **Endogeneity:** Predictors such as `minutes_played` are likely endogenous (better players get more minutes). The coefficient for this variable should be interpreted as an association reflecting "Squad Status" rather than a purely causal effect of playing time.

	Model_2.3	Model_3.2	Model_4
(Intercept)	15.993 ***	16.850 ***	16.857 ***
	(0.076)	(0.078)	(0.079)
poly(season_year_c, 3, raw = TRUE)1	-0.093 ***	-0.056 **	-0.061 ***
	(0.025)	(0.017)	(0.017)
poly(season_year_c, 3, raw = TRUE)2	0.022 ***	0.028 ***	0.028 ***
	(0.005)	(0.003)	(0.003)
poly(season_year_c, 3, raw = TRUE)3	-0.002 ***	-0.002 ***	-0.002 ***
	(0.000)	(0.000)	(0.000)
height_c		0.010 **	0.010 **
		(0.004)	(0.004)
positionDefender		-0.404 ***	-0.404 ***
		(0.058)	(0.058)
positionGoalkeeper		-1.057 ***	-1.057 ***
		(0.095)	(0.095)
positionMidfield		-0.190 ***	-0.190 ***
		(0.058)	(0.058)
citizen_of_club_countryYes		-0.190 ***	-0.186 ***
		(0.031)	(0.031)
RegionEast Asia & Pacific		-0.431 *	-0.432 *
		(0.204)	(0.204)
RegionLatin America & Caribbean		0.191 **	0.189 **
		(0.061)	(0.061)
RegionMiddle East & North Africa		-0.092	-0.094
		(0.237)	(0.237)
RegionNorth America		-0.063	-0.067
		(0.217)	(0.217)
RegionSub-Saharan Africa		0.066	0.064
		(0.100)	(0.100)
poly(age_c, 2, raw = TRUE)1		0.019 ***	0.018 ***
		(0.004)	(0.004)
poly(age_c, 2, raw = TRUE)2		-0.024 ***	-0.024 ***